

LAKEVIEW UNIVERSITY

Department of Computer Science

CS 211 - INTRODUCTION TO SYSTEMS

Spring 2026 | 3 Credit Hours | CRN: 71195

Instructor: Prof. James Hartley

Office: Tech Center 215

Office Hours: Monday 2:00 - 4:00 PM, Wednesday 10:00 AM - 12:00 PM

Email: jhartley@lakeviewu.edu

Meeting Times: Tuesday, Thursday 9:30 - 10:45 AM

Location: Tech Center 208

COURSE DESCRIPTION

This course introduces computer systems from the perspective of a programmer. Topics include data representation, machine-level programming, the memory hierarchy, processes, and basic operating system concepts. Students learn to read and write C code and x86-64 assembly, use debugging tools, and understand how high-level programs map to hardware. Lab sessions provide hands-on practice on Linux systems.

Prerequisites: CS 102 with a grade of C or better.

REQUIRED MATERIALS

- **Textbook:** Computer Systems: A Programmer's Perspective, 3rd Edition, Bryant and O'Hallaron (CS: APP) - available on reserve at Lakeview Library
- GCC toolchain (available on lab machines; SSH access provided)
- Access to Lakeview course portal

GRADING

Component	Weight
Labs (7 total)	40%
Midterm Exam	25%
Final Exam	30%
Attendance	5%

Grading Scale: Same as department standard (see CS 201 syllabus format)

COURSE SCHEDULE

Week Topic

- 1 Course overview, bits and bytes, integer representation
 - 2 Integer arithmetic, overflow, two's complement
 - 3 Floating point representation (IEEE 754)
 - 4 Machine-level programming: basics, registers, operands
 - 5 Machine-level programming: control, conditionals, loops
 - 6 Procedures: stack frames, calling conventions
 - 7 Data structures in assembly: arrays, structs, unions
 - 8 MIDTERM EXAM (in class, date posted to portal)
 - 9 Memory hierarchy: caches, locality
 - 10 Cache performance, memory mountain

 - 11 Linking: static and dynamic, libraries
 - 12 Processes and signals: fork, exec, wait
 - 13 Virtual memory: address spaces, page tables
 - 14 Exceptions and system calls
 - 15 Final review
- Finals FINAL EXAM (date set by registrar)
-

LABS

- Lab 1 Data Lab:** manipulating integers and floats with bitwise operations
- Lab 2 Bomb Lab:** defuse a binary bomb by reverse engineering assembly
- Lab 3 Attack Lab:** exploit buffer overflows
- Lab 4 Cache Lab:** write a cache simulator and optimize matrix code
- Lab 5 Shell Lab:** implement a Unix shell with job control
- Lab 6 Malloc Lab:** implement a dynamic memory allocator
- Lab 7 Proxy Lab:** implement a concurrent web proxy

Labs are submitted via the course portal. Late labs lose 15% per 24 hours.
Working with a partner on labs is not permitted unless explicitly stated.

USE OF TOOLS AND ACADEMIC INTEGRITY

Debugging tools (gdb, valgrind) are encouraged. Sharing lab solutions or using solutions from previous semesters is a violation of academic integrity. This applies to all posted, shared, or AI-generated code.

ACCOMMODATIONS

Contact Disability Services and notify the instructor within the first two weeks for any accommodations.

Syllabus subject to minor revision. Check the course portal for updates.